

32°C Light rain Search [Taskbar icons: File Explorer, Edge, Word, etc.] ENG IN 10:31 13-08-2026

MATLAB

sriyukthesh100-5G6GLab/5G-6G

MIMO Simulation Fix

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Ask Gemini

matlab.mathworks.com

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FIGURE

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Search (Ctrl+Shift+Space)

Save As

Copy Figure

Print

Show Code

FILE

Labels and Annotations

Title

Subtitle

X-Label

Y-Label

Z-Label

Legend

Colorbar

Grid

X-Grid

Y-Grid

MATLAB Drive

Files

SlotC_Exp2b.m

SlotC_Exp2c.m

SlotC_Exp3.m

SlotC_Exp4a.m

SlotC_Exp4b.m

SlotC_Exp4c.m

SlotC_Exp5a.m

SlotC_Exp5b.m

SlotC_Exp5c.m

SlotC_Exp6a.m

SlotC_Exp6b.m

Workspace

Name	Value	Size	Class
antennas	[2,4,8,16,32]	1×5	double
gain	[3.0103,6.020...	1×5	double

SlotC_Exp6a.m

SlotC_Exp6b.m

SlotC_Exp6c.m

SlotC_Exp7a.m

Figure 1

Beamforming Gain for Different Antenna Array Sizes

Number of Antenna Elements	Beamforming Gain (dB)
2	3.01
4	6.02
8	9.03
16	12.04
32	15.05

% Number of Antenna Elements

antennas = [2 4 8 16 32];

% Beamforming Gain (dB)

gain = 10 * log10(antennas);

figure

% Bar graph

bar(antennas, gain)

grid on

xlabel('Number of Antenna Elements')

ylabel('Beamforming Gain (dB)')

Command Window

New to MATLAB? See resources for [Getting Started](#).

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